

Bhargav Amin

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Summary

Data Science Master’s student and Computer Science (AIML) graduate with 1+ years of experience in predictive modeling and full-stack development. Specialist in end-to-end data pipelines, time-series forecasting, and AI automation.

Education

- The University of Texas at Arlington** Arlington, Texas
Masters in Data Science Jan 2025 - Present
- PP Savani University** Gujarat, India
Bachelor of Technology: Computer Science Engineering (AIML) Aug 2020 - May 2024

Experience

- Enlighten Infosystems - Vadodara, India | Data Science Intern** Dec 2023 – Dec 2024
 - Built and compared multiple time-series forecasting models (TimesFM, Prophet, PatchTST) to improve demand predictions, contributing to a 3% increase in sales.
 - Designed and deployed an automated ML pipeline using GCP Vertex AI, Docker, and Azure REST APIs, reducing model retraining and deployment time by 40%.
 - Developed reliable data pipelines and dashboards using Apache NiFi, Airflow, Superset, and Tableau to support large-scale ETL, monitoring, and business decision-making.
- Bodhivriksh Bizsol LLP - Vadodara, India | Web Intern** Jun 2022 - Jul 2022
 - Built a full-stack HR management web application using PHP and Flask to automate internal workflows and simplify employee data management.
 - Worked closely with HR and IT teams to gather requirements and deliver fully integrated features aligned with business needs.
 - Designed responsive, user-friendly interfaces and backend integrations using HTML, CSS, JavaScript, and MySQL to enable secure, real-time HR reporting.

Projects

- Image Caption Generator using InceptionV3 and Transformer** [🔗](#)
 - Developed an image captioning model using InceptionV3 as the encoder and a Transformer-based decoder on the Flickr8k dataset, generating context-aware captions from images.
 - Achieved BLEU Score: 4.2% and METEOR Score: 18.8%, reflecting the model’s captioning performance and semantic alignment.
 - Technologies used:** Python, TensorFlow/Keras, InceptionV3, Transformer, NLP, Computer Vision
- SandStorm Semantic Segmentation Model using U-Net Architecture** [🔗](#)
 - Developed a deep learning model based on U-Net architecture that segments sandstorm images into 4 distinct intensity layers, enabling accurate visualization and differentiation between high and low storm areas for meteorological analysis
 - Trained the model on a specialized SandStorm dataset to achieve precise pixel-level segmentation, providing real-time storm intensity assessment and geographic impact visualization capabilities
 - Technologies used:** Python, TensorFlow, U-Net Architecture, OpenCV
- Corrective-RAG- Adaptive RAG with Query Rewriting** [🔗](#)
 - Achieved a 20% increase in relevant document retrieval by integrating Chroma as a vector store and BAAI/bge-small-en-v1.5 embeddings using HuggingFace for semantic search.
 - Improved user experience by incorporating query rewriting and fallback web search via DuckDuckGo when retrieval confidence was low, enhancing overall response quality.
 - Technologies used:** Python, HuggingFace, ChromaDB, LangGraph, NLP, and DuckDuckGo Search API.

Accomplishments

- Co-authored a research paper on data science methodologies while at Enlighten Infosystems, published in collaboration with the company’s founder and a colleague, contributing to industry knowledge and enhancing our professional credibility. [🔗](#)
- Website Development for University TechFest event Tech-Pulse (2022)
- Got 5th rank in UTA Datathon (2025) for building best forecasting model

Certifications

- Deep Learning from Coursera (2024)
- Data Science from Coursera IBM (2024)
- Google Professional Data Analytics from Coursera(2023)
- The Complete Tableau Bootcamp Data Visualization from Udemy
- (2022) Data Science Real World Projects in Python from Udemy (2021)

Skills

- Languages:** C++, Python, SQL, MySql, SAS
- Libraries:** Numpy, pandas, Matplotlib, plotly, Sklearn, Tensorflow, OpenCV, scikit-learn
- Image Processing:** Object detection, Image classification, Image segmentation
- GenAI Tools:** OpenAI Api, Azure AI Foundry,
- Big Data And Cloud:** Microsoft Azure, Google Cloud Platform, Apache Spark
- Technologies:** Docker, Power BI, Flask, Streamlit, Airflow, Nifi, Android Development (Java), BeautifulSoup